

RideShield

UPI-AutoPay Dynamic Micro-Insurance for Gig Workers

Feasibility tag: Buildable in 36–48 hrs · **Category:** Software · **Theme:** FinTech / InsurTech / Social Welfare

Rubric Weight Map

#	Rubric Criterion	What it's really testing	Marks
1	Problem Statement Understanding & Relevance	Do you understand who's affected and why now?	10
2	Problem Analysis & Root Cause	Have you found the specific mismatch, not a generic one?	7
3	Literature Survey & Existing Solutions	Did you find the real adjacent products, or did the mentor have to?	8
4	Valid Citations & Evidence	Traceable, not invented	5
5	Proposed Idea & Problem–Solution Fit	Solution maps to the specific unsolved slice	10
6	Innovation / Differentiation	Can you survive “isn't this already a product?”	5
7	Feasibility & Expected Impact	Honest numbers, honest scope	5

1. Problem Statement Understanding & Relevance (10 marks)

One-line PS: “Design a behavioural, usage-linked micro-insurance system that gives India's gig delivery/ride workers real accident coverage priced and paid per shift, not per month.”

Stakeholders: gig riders (Zomato / Swiggy / Rapido / Uber / Ola), IRDAI-regulated micro-insurers, NPCI (UPI AutoPay rails), platforms, state labour departments.

Why now — lead with the strongest, most current evidence

- India's Economic Survey 2025–26 records gig-sector workers growing from 7.7 million in FY2021 to 12 million in FY2025, alongside smartphone penetration crossing 800 million users and roughly 15 billion monthly UPI transactions — and states plainly that gig workers still face income volatility that limits financial inclusion.
- A 2026 field study on India's gig platforms found 24% of drivers cited road accidents as a safety concern and explicitly recommended crash-detection features.
- Research into gig-worker vehicle insurance found more than half of two-wheeler and cab-based gig workers in India carry no insurance coverage at all, despite state mandates requiring third-party motor cover.
- Policy tailwind: the Code on Social Security, 2020 already formally recognises gig and platform workers for social security schemes; Karnataka's 2023 budget committed a fully state-funded ₹4 lakh insurance package for e-commerce delivery workers.

This section defends directly against the low-score bullet: “Problem statement is broad, vague or not supported by evidence.”

2. Problem Analysis & Root Cause (7 marks)

Go one level past “gig workers lack insurance” — show the mechanism:

- Income-structure mismatch:** gig income is daily and volatile, but almost every existing insurance product assumes a stable monthly premium built for salaried individuals or fleet owners.
- Comprehension / procedural barrier, not eligibility:** a 2026 data study on Indian gig workers found low pension/insurance participation stems mainly from limited information and procedural friction, not ineligibility.

Root-cause conclusion to state on your slide: *Gig riders aren't uninsured because they don't want cover — they're uninsured because every existing product, platform group cover included, is priced and administered for a stable, single-employer relationship they don't have. RideShield doesn't sell insurance differently; it re-times and re-prices the same protection to match how gig income actually arrives: daily, volatile, and shift-based.*

This section defends directly against: “Solution is presented without explaining the actual problem or user need.”

3. Literature Survey & Existing Solutions (8 marks)

Existing approach	What it does	Where it falls short
Platform group-accident cover (Zomato, Swiggy, Ola, Uber, Rapido — via Acko, Reliance General)	Free or subsidised personal-accident/health cover bundled with platform membership — e.g. Swiggy's ₹12 lakh cover via Acko	Flat-rate and employer-funded, not priced by actual on-road risk; coverage has been reported to lapse when a rider falls short of platform engagement quotas; claims remain a manual, paperwork-based process
PAYD / PHYD telematics motor insurance (Acko, Digit, HDFC ERGO, Tata AIG, ICICI Lombard, Bajaj Allianz, under IRDAI's 2022 usage-based add-on circular)	GPS/OBD-based pricing of vehicle own-damage cover by distance driven (PAYD) or driving style (PHYD)	Built for private car owners on an annual/km-slab policy term, not per-shift micro-premiums; covers the vehicle, not the rider's personal injury; no gig-worker or two-wheeler-delivery variant in market yet
Weather-triggered parametric cover for Indian gig/informal workers (Bajaj Allianz "ClimateSafe," Apr 2025; Go Digit × Jan Sahas heat-index pilots, Delhi NCR, 2025–26; SEWA parametric heat cover)	Automatic payout when a weather index (heat, AQI, rainfall) crosses a threshold — no claim form, no proof of loss required	Proves the "no-paperwork parametric payout" model works for Indian gig workers, but the trigger is an external weather/AQI index, not the accident itself — it compensates income lost to conditions, not a crash
Early-stage / hackathon-brief concepts for AI-parametric gig-delivery insurance	AI-estimated, trigger-based payouts for gig-worker income protection, several explicitly scoped to weather/disruption triggers	Pre-product; at least one such brief explicitly excludes accident, health, life, and vehicle-repair cover from scope — cite the category honestly, not a specific unverifiable claim
Regulatory framework (IRDAI Micro Insurance Regulations, 2015; IRDAI's 2022 usage-based motor add-on circular; Code on Social Security 2020, Chapter IX, in force 21 Nov 2025)	Permits small-ticket bundled products and usage-based pricing; mandates aggregator contribution (1–2% of turnover, capped at 5% of worker payments) to a Social Security Fund covering gig workers	Enables everything RideShield proposes but doesn't itself build it — no notified product yet combines usage-based pricing with the Fund mechanism for accident cover specifically

Literature gap to state precisely: *Usage-based pricing is a proven, IRDAI-permitted category (telematics motor insurance). No-paperwork parametric payouts are a proven, actively-piloted category for Indian gig and informal workers (heat/AQI parametric cover). Nobody has combined the two for accident risk specifically: existing parametric pilots for Indian gig workers trigger on weather, not on the crash itself, and existing telematics UBI prices private car owners by the policy year, not two-wheeler delivery riders by the shift. That combination — plus a premium small enough to come out of a single day's earnings — is the open gap RideShield sits in.*

COMPETITOR / NAMING FLAG

A 2026 hackathon brief circulating under a similar description ("AI-powered parametric insurance for gig delivery partners") targets weather/disruption-linked income loss and explicitly excludes accident cover from scope. Raise this yourself if a mentor asks "isn't this the same brief" — RideShield's trigger is the crash itself (accelerometer/gyroscope + GPS-hospital match), not a weather index, and the premium mechanism (UPI AutoPay, per-shift) differs from a weekly income-protection pricing model. Name the distinction yourself before it's asked.

Defends against: "Literature is listed but not compared to identify a meaningful gap."

4. Valid Citations & Evidence (5 marks)

- Economic Survey 2025–26 (gig-workforce growth, 7.7M FY2021 → 12M FY2025; UPI transaction volume)
- 2026 field study on Indian gig-platform drivers (road-accident safety concern, 24% figure)
- Research on gig-worker two-wheeler/cab insurance coverage gap (uninsured-rider figure)
- 2026 data study on Indian gig workers' pension/insurance participation barriers
- Code on Social Security, 2020, Chapter IX (in force 21 Nov 2025) — aggregator Social Security Fund contribution
- IRDAI (Micro-insurance) Regulations, 2015
- IRDAI circular permitting Pay-As-You-Drive / Pay-How-You-Drive motor own-damage add-on covers (2022)
- Press coverage of Zomato/Swiggy rider-insurance partnerships with Acko and Reliance General
- Bajaj Allianz "ClimateSafe" gig-worker parametric weather insurance launch, April 2025 (industry press)
- Go Digit × Jan Sahas heat-parametric insurance pilot for informal/gig workers, Delhi NCR, 2025–26 (industry & policy press)
- Karnataka state budget 2023 — e-commerce delivery worker insurance package

Rule for your team: anything you can't trace to one of these gets cut or rewritten as a hedged, defensible claim — the same discipline applied to MuleTrace and GhostLock's reference lists.

5. Proposed Idea & Problem–Solution Fit (10 marks)

Root cause (Section 2 / 3)	RideShield mechanism
Fixed monthly premium mismatched to daily, volatile gig income	UPI AutoPay micro-premium of ₹2–5 deducted only on days the rider logs in to work — priced and paid per shift, not per month
Low participation from comprehension/procedural friction, not ineligibility	Passive, zero-form enrolment — coverage activates automatically the moment a rider logs in for a shift; no application, no underwriting questionnaire
Existing platform cover is flat-rate regardless of actual on-road risk, and can lapse when engagement quotas aren't met	On-device telemetry risk engine reprices the premium per shift from real driving conditions (speed, braking, weather), so cost reflects actual exposure, not platform-tenure quotas
Existing parametric pilots for Indian gig workers trigger on weather, not on the accident itself	Parametric trigger tuned specifically to crash-signature telemetry (accelerometer/gyroscope + stationary check) or a GPS-hospital location match — payout tied to the accident event itself
Claims friction	Gyroscope crash-signature + GPS-hospital match auto-initiates the claim

Defends against: "Idea is essentially a generic chatbot/app/dashboard without differentiation."

6. Innovation / Differentiation (5 marks)

State this as one crisp line for the slide: *"Unlike static state welfare schemes (enrollment-based) or existing flat-rate micro-insurance pilots (fixed pricing), RideShield is the first system to combine live behavioral risk pricing with automatic parametric claims — on a consumer smartphone, with no dedicated hardware."*

7. Feasibility & Expected Impact (5 marks) — this is where your gap doc becomes your biggest asset

Your mentor will almost certainly probe: *"What happens when a pothole triggers a false claim?"* and *"How do you actually price this?"* — your research gap document already identified these as Gap 1 and Gap 2. Don't hide from them — present them as **known challenges with a scoped mitigation**, which is exactly what separates a mature team from a naive one in a mentor's eyes.

Challenge 1 — False-positive crash detection (from your gap doc: even a 5% false-positive rate threatens insurability)

- *Mitigation for hackathon scope:* multi-signal confirmation — accelerometer + gyroscope + a stationary-time window, not a single sensor spike — filters out potholes/hard braking from genuine crashes. Add a **human-in-the-loop WhatsApp confirmation step** ("We detected a possible accident — are you okay? Tap to confirm.") before any payout fires, instead of a fully-automatic trigger. This removes most of the paperwork friction while keeping a safety check.
- *What you say to your mentor:* "We know full field validation needs 500–1,000 riders over months — that's future work. For the hackathon, our MVP demonstrates the rule-based filtering logic on simulated road-event data (pothole vs. crash signatures) to show the discrimination approach works in principle."

Challenge 2 — Actuarial pricing / adverse selection (Gap 2 in your doc: no published pricing model exists for this population)

- *Mitigation:* don't position RideShield as *becoming* the insurer — position it as a **risk-scoring and collection layer that plugs into an existing IRDAI-licensed micro-insurance partner**, since ₹150–200/month accident products already exist in-market. This sidesteps inventing actuarial science in 36 hours and reframes your innovation honestly as the *distribution and real-time-pricing layer*, not the underwriting itself.

Scope honestly — what's mocked vs. built live:

- Built live: telemetry risk engine, UPI AutoPay flow (sandbox), WhatsApp bot claim conversation, crash-signature filtering logic
- Explicitly mocked and labeled as such: actual underwriting/payout settlement, insurer backend

Expected impact: even modest adoption against a 12-million-strong and growing gig workforce (FY25) represents meaningful coverage-gap closure, with the Code on Social Security 2020 giving a plausible path to scale beyond a single-state pilot like Karnataka's.

Defends against: "Scope is unrealistic for the available time/resources" and "Claims of novelty, accuracy or impact are unsupported."

Final pre-submission checklist

Low-score trap	RideShield's defense
Vague/unsupported problem	Section 1 evidence chain
Solution without user	Section 2 root-cause mapping

need	
No literature survey	Section 3 comparison table
Fabricated/irrelevant references	Section 4 fixed reference list
Generic app without differentiation	Section 6 one-line differentiator
Unrealistic scope	Section 7 explicit mock-vs-live split
Unsupported novelty claims	Every claim traced to Section 4 sources
Names/institution in PPT/video	Strip all names, dept, and college identifiers from every slide and demo video now, as a habit — even for the mentor round, since this deck will likely get reused for the judged round later